The proof of (iii) is a trivial consequence of (ii) and the summary in remark 11.13. ■

We won't bother to discuss the case $\xi \geq \frac{1}{2}$.

12. Apriori bounds for $\varphi(s)$, $E(z;s;\chi)$, $\varphi_m(s)$.

Before proving the trace formula, we need to develop certain estimates for $E(z;s;\chi)$ when $\mathrm{Re}(s) \geq \frac{1}{2}$. Cf. page 82 line 3. The material in this section will correspond to Selberg[2,pp.52-64].

Review assumptions (i)-(ix) in section 3. Take $\chi(S) = 1$, and recall proposition 8.6. See also theorem 11.8(F). As in section 11, we define:

(12.1) $$B = 5 + y_0 \quad , \quad y_0 \geq 1000 .$$

LEMMA 12.1. For $\frac{1}{2} \leq \mathrm{Re}(s) \leq \frac{3}{2}$ and $y > 0$, we have:

$$|K_{s-\frac{1}{2}}(y)| \leq \frac{3e^{-y}}{\sqrt{y}}\left(1 + \frac{1}{\sqrt{y}}\right) .$$

Proof. Write $s = \sigma + it$ and refer to lemma 4.8(iii). We immediately obtain:

$$K_{s-\frac{1}{2}}(y) = \int_0^\infty e^{-y\cosh\xi} \cosh[(s-\tfrac{1}{2})\xi]\, d\xi$$

$$K_{s-\frac{1}{2}}(y) = \frac{1}{2}\int_0^\infty e^{-y\cosh\xi} [e^{(s-\frac{1}{2})\xi} + e^{(\frac{1}{2}-s)\xi}]\, d\xi$$

$$|K_{s-\frac{1}{2}}(y)| \leq \frac{1}{2}\int_0^\infty e^{-y\cosh\xi} [e^{(\sigma-\frac{1}{2})\xi} + e^{(\frac{1}{2}-\sigma)\xi}]\, d\xi$$

$$\{ v = e^{\xi} , \xi = \ln v \}$$

$$|K_{s-\frac{1}{2}}(y)| \leq \frac{1}{2}\int_1^\infty e^{-\frac{y}{2}(v+v^{-1})} [v^{\sigma-\frac{1}{2}} + v^{\frac{1}{2}-\sigma}] \frac{dv}{v}$$

$$|K_{s-\frac{1}{2}}(y)| \leq \frac{1}{2} e^{-y} \int_1^\infty e^{-\frac{y}{2}(\sqrt{v} - \frac{1}{\sqrt{v}})^2} [v^{\sigma-\frac{1}{2}} + v^{\frac{1}{2}-\sigma}] \frac{dv}{v}$$

$$\leq e^{-y} \int_1^\infty e^{-\frac{y}{2}(\sqrt{v}-1)^2} v^{\sigma-\frac{1}{2}} \frac{dv}{v}$$

$$\leq e^{-y} \int_1^\infty e^{-\frac{y}{2}(\sqrt{v}-1)^2} dv$$

$$\{ w = \sqrt{v} - 1 , \quad v = (w+1)^2 \}$$

$$= e^{-y} \int_0^\infty e^{-\frac{y}{2}w^2} 2(w+1)\, dw$$

$$= 2e^{-y} \int_0^\infty e^{-\frac{y}{2}w^2} w\, dw + 2e^{-y} \int_0^\infty e^{-\frac{y}{2}w^2} dw$$

$$= 2e^{-y} \int_0^\infty e^{-yu} du + 2e^{-y} \sqrt{\frac{\pi}{2y}} \qquad \text{BMP}[4, p. 15(11)]$$

$$= 2\frac{e^{-y}}{y} + \sqrt{2\pi}\, \frac{e^{-y}}{\sqrt{y}}$$

$$\leq 3\frac{e^{-y}}{\sqrt{y}} \left(1 + \frac{1}{\sqrt{y}}\right) . \qquad \blacksquare$$

LEMMA 12.2. Suppose that $s = \frac{1}{2} + h + it$, $0 \leq h \leq 1$, $t \in \mathbb{R}$, $\eta > 0$. Then:

$$\int_\eta^\infty |K_{s-\frac{1}{2}}(y)|^2 \frac{dy}{y} \geq A e^{-5\eta - 5|t|} \qquad \boxplus$$

where A is some [positive] constant.

Proof. To begin with, we assume that $\eta > 1$. The basic idea is to exploit

$$\int_\eta^T |K_{s-\frac12}(y)|^2 \frac{dy}{y} \geqq \frac{\left(\int_\eta^T |f(y)K_{s-\frac12}(y)|\,dy\right)^2}{\int_\eta^T |f(y)|^2 y\,dy}$$

for <u>appropriate</u> T and f . The simplest choice (☆) of f is suggested by the Sonine-Gegenbauer formula:

$$\int_0^\infty \frac{K_\nu[\sqrt{\xi^2+x^2}\,]}{(\xi^2+x^2)^{\nu/2}}\,\xi^{2\mu+1}\,d\xi = 2^\mu\,\Gamma(1+\mu)\,x^{1+\mu-\nu}K_{\nu-\mu-1}(x) \quad .$$

Here $x > 0$ and $\mathrm{Re}(\mu) > -1$. Cf. BMP[2,p.95(50)], MOS[1,p.104(bottom)], and Watson[1,p.417(6)]. Take $\nu = h + it$, $\mu = it$, $x = \eta$. At once:

$$\int_\eta^\infty K_{s-\frac12}(y)\,y^{1-h-it}(y^2-\eta^2)^{it}\,dy = 2^{it}\,\Gamma(1+it)\,\eta^{1-h}K_{h-1}(\eta) \quad .$$

Consult lemma 4.8(iii) for $K_{h-1}(\eta)$. Therefore:

$$|\Gamma(1+it)|\,\eta^{1-h}K_{h-1}(\eta) \leqq \int_\eta^\infty |K_{s-\frac12}(y)|\,y^{1-h}\,dy$$

$$|\Gamma(1+it)|\,\eta^{1-h}K_0(\eta) \leqq \int_\eta^\infty |K_{s-\frac12}(y)|\,y^{1-h}\,dy$$

$$\left\{\begin{array}{c} \text{but } K_0(\eta) \geqq c_1\dfrac{e^{-\eta}}{\eta} \quad \text{and} \quad |\Gamma(1+it)| \geqq c_2 e^{-\frac{\pi}{2}|t|} \\ \hline \text{with absolute constants } c_j > 0 \\ \hline \text{cf. lemma 4.8(iv) and BMP [1, page 47(6)]} \end{array}\right\}$$

(☆) See remark 12.3.

$$c_3 e^{-\frac{\pi}{2}|t|} \eta^{-h} e^{-\eta} \lessapprox \int_\eta^\infty |K_{s-\frac{1}{2}}(y)| y^{1-h} dy$$

$$c_3 e^{-\frac{\pi}{2}|t|} \eta^{-h} e^{-\eta} - \int_T^\infty |K_{s-\frac{1}{2}}(y)| y^{1-h} dy \lessapprox \int_\eta^T |K_{s-\frac{1}{2}}(y)| y^{1-h} dy$$

{ take $T \gtrapprox 1+\eta$; exploit lemma 12.1 }

$$c_3 e^{-\frac{\pi}{2}|t|} \eta^{-h} e^{-\eta} - 6 \int_T^\infty e^{-y} y^{\frac{1}{2}-h} dy \lessapprox \int_\eta^T |K_{s-\frac{1}{2}}(y)| y^{1-h} dy$$

$$c_3 e^{-\frac{\pi}{2}|t|} \eta^{-h} e^{-\eta} - 6 \int_T^\infty e^{-y} y^{\frac{1}{2}} dy \lessapprox \int_\eta^T |K_{s-\frac{1}{2}}(y)| y^{1-h} dy$$

{recall BMP[1, p. 266(21)] and BMP[2, p. 135(6)] }

$$c_3 e^{-\frac{\pi}{2}|t|} \eta^{-h} e^{-\eta} - c_4 T^{\frac{1}{2}} e^{-T} \lessapprox \int_\eta^T |K_{s-\frac{1}{2}}(y)| y^{1-h} dy$$

$$c_3 \frac{e^{-\frac{\pi}{2}(\eta+|t|)}}{(\eta+|t|)^h} - c_4 T^{\frac{1}{2}} e^{-T} \lessapprox \int_\eta^T |K_{s-\frac{1}{2}}(y)| y^{1-h} dy$$

$$\left\{ \begin{array}{c} \text{take } T = 2(\eta+|t|+c_5) \text{ with } c_5 \approx \infty \\ \hline \text{remember that } 0 \leqq h \leqq 1 \end{array} \right\}$$

$$\frac{1}{2} c_3 \left[\frac{e^{-\frac{\pi}{2}(\eta+|t|+c_5)}}{(\eta+|t|+c_5)^h} \right] \lessapprox \int_\eta^T |K_{s-\frac{1}{2}}(y)| y^{1-h} dy$$

$$c_6 e^{-2(\eta+|t|+c_5)} \lessapprox \int_\eta^T |K_{s-\frac{1}{2}}(y)| y^{1-h} dy \quad .$$

Consequently:

$$\int_\eta^T |K_{\lambda+it}(y)|^2 \frac{dy}{y} \geq \frac{[c_6 e^{-2(\eta+|t|+c_5)}]^2}{\int_\eta^T y^{3-2\lambda}\,dy} \geq c_7 \frac{e^{-4(\eta+|t|+c_5)}}{T^4}$$

$$\int_\eta^\infty |K_{\lambda+it}(y)|^2 \frac{dy}{y} \geq c_8 e^{-5(\eta+|t|+c_5)} = c_9 e^{-5\eta-5|t|} .$$

This estimate presupposes that $\eta > 1$. To treat $\eta > 0$, we merely observe that

$$\int_\eta^\infty |K_{s-\frac{1}{2}}(y)|^2 \frac{dy}{y} \geq \int_{1+\eta}^\infty |K_{s-\frac{1}{2}}(y)|^2 \frac{dy}{y} \geq c_9 e^{-5\eta-5-5|t|} .$$

Thus $A = c_9 e^{-5}$. ■

REMARK 12.3. To prove lemma 12.2, we had to select an f so that

$$\int_\eta^\infty f(y) K_{\lambda+it}(y)\,dy$$

is easily estimated. Since both t and y can become large, the familiar asymptotic relation for $K_{s-\frac{1}{2}}(y)$ runs into [serious] trouble. Cf. BMP[2,pp.22-31] and lemma 4.8(iv). The Sonine-Gegenbauer formula avoids this problem by allowing us to transform $K_{s-\frac{1}{2}}(y)$ into $K_{h-1}(\eta)\Gamma(1+it)$.

Observe that the number of candidates for f is very small. See GR[1,pp.665-775].

PROPOSITION 12.4. The analytic function $\varphi(s)$ is uniformly bounded whenever $\frac{1}{2} \leq \mathrm{Re}(s) \leq \frac{3}{2}$ and $|\mathrm{Im}(s)| \geq 1$. In this regard, see proposition 8.6(d).

Proof. Write $s = \frac{1}{2} + h + it$. WLOG $0 < h \leq 1$, $|t| \geq 1$. Cf. remark 8.3 and theorem 11.8. For $\mathrm{Re}(\xi) \geq \frac{1}{2}$, we define:

$$E_0(z;\xi;\%) = \begin{cases} E(z;\xi;\%) & \text{when} \quad z \in \operatorname{int}(\mathcal{F}_B) \\ E(z;\xi;\%) - y^{\xi} - \varphi(\xi) y^{1-\xi} & \text{when} \quad z \in \operatorname{int}(\mathcal{C}_B) \end{cases} .$$

Recall equations (11.1) - (11.5). Take $u = v = E_0(z;s;\%)$ in Green's identity. Then:

$$\iint\limits_{\mathcal{F}_B} [u \Delta \bar{v} - \bar{v} \Delta u] \, dx\, dy = \int\limits_{\partial \mathcal{F}_B} \left[u \frac{\partial \bar{v}}{\partial n} - \bar{v} \frac{\partial u}{\partial n}\right] |dz|$$

$$\iint\limits_{\mathcal{F}_B} [u D(\bar{v}) - \bar{v} D(u)] \, d\mu(z) = \int\limits_{\mathcal{L}_B} \left[u \frac{\partial \bar{v}}{\partial n} - \bar{v} \frac{\partial u}{\partial n}\right] |dz|$$

$$[s(1-s) - \bar{s}(1-\bar{s})] \iint\limits_{\mathcal{F}_B} u \bar{v} \, d\mu(z) = \int\limits_{\mathcal{L}_B^-} \left[u \frac{\partial \bar{v}}{\partial y} - \bar{v} \frac{\partial u}{\partial y}\right] |dz|$$

{ cf. page 110 for $\mathcal{L}_B^{\pm}$ }

$$[s(1-s) - \bar{s}(1-\bar{s})] \iint\limits_{\mathcal{C}_{B,Y}} u \bar{v} \, d\mu(z) = \int\limits_{\mathcal{L}_Y} \left[u \frac{\partial \bar{v}}{\partial y} - \bar{v} \frac{\partial u}{\partial y}\right] |dz| - \int\limits_{\mathcal{L}_B^+} \left[u \frac{\partial \bar{v}}{\partial y} - \bar{v} \frac{\partial u}{\partial y}\right] |dz|$$

$$[s(1-s) - \bar{s}(1-\bar{s})] \iint\limits_{\mathcal{F}_Y} |u|^2 \, d\mu(z) = \int\limits_{\mathcal{L}_B^-} \left[u \frac{\partial \bar{v}}{\partial y} - \bar{v} \frac{\partial u}{\partial y}\right] |dz| + \int\limits_{\mathcal{L}_Y} - \int\limits_{\mathcal{L}_B^+}$$

{ apply theorem 11.8 (F) and Parseval's equation
- - - - - - - - - - - - - - - - - - - -
cf. the proof of claim 9.6 }

$$[s(1-s)-\bar{s}(1-\bar{s})] \iint_{\mathcal{F}_Y} |u|^2 d\mu(z) = [\mathcal{B}^s + \varphi(s)\mathcal{B}^{1-s}]\, \overline{[s\mathcal{B}^{s-1} + (1-s)\varphi(s)\mathcal{B}^{-s}]}$$

$$- \overline{[\mathcal{B}^s + \varphi(s)\mathcal{B}^{1-s}]}\, [s\mathcal{B}^{s-1} + (1-s)\varphi(s)\mathcal{B}^{-s}]$$

$$+ \sum_{m \neq 0} \varphi_m(s)\overline{\varphi_m(s)}\, Y^{\frac{1}{2}} K_{s-\frac{1}{2}}(2\pi|m|Y) \frac{\partial}{\partial Y}[Y^{\frac{1}{2}} K_{\bar{s}-\frac{1}{2}}(2\pi|m|Y)]$$

$$- \sum_{m \neq 0} \varphi_m(s)\overline{\varphi_m(s)}\, Y^{\frac{1}{2}} K_{\bar{s}-\frac{1}{2}}(2\pi|m|Y) \frac{\partial}{\partial Y}[Y^{\frac{1}{2}} K_{s-\frac{1}{2}}(2\pi|m|Y)]$$

$$(s-\bar{s})(1-s-\bar{s}) \iint_{\mathcal{F}_Y} |u|^2 d\mu(z) = (\bar{s}-s)\mathcal{B}^{s+\bar{s}-1} + \varphi(s)(s+\bar{s}-1)\mathcal{B}^{\bar{s}-s} + \overline{\varphi(s)}(1-s-\bar{s})\mathcal{B}^{s-\bar{s}} + |\varphi(s)|^2(s-\bar{s})\mathcal{B}^{1-s-\bar{s}}$$

$$+ O(e^{-4\pi Y})$$

{ the implied constant may depend upon s }

$$(s-\bar{s})(1-s-\bar{s}) \iint_{\mathcal{F}} |u|^2 d\mu(z) = (\bar{s}-s)\mathcal{B}^{s+\bar{s}-1} + \varphi(s)(s+\bar{s}-1)\mathcal{B}^{\bar{s}-s} + \overline{\varphi(s)}(1-s-\bar{s})\mathcal{B}^{s-\bar{s}} + |\varphi(s)|^2(s-\bar{s})\mathcal{B}^{1-s-\bar{s}}$$

$$\iint_{\mathcal{F}} |E_0(z;s;\chi)|^2 d\mu(z) = \frac{\mathcal{B}^{s+\bar{s}-1}}{s+\bar{s}-1} + \varphi(s)\frac{\mathcal{B}^{\bar{s}-s}}{\bar{s}-s} + \overline{\varphi(s)}\frac{\mathcal{B}^{s-\bar{s}}}{s-\bar{s}} + |\varphi(s)|^2 \frac{\mathcal{B}^{1-s-\bar{s}}}{1-s-\bar{s}}$$

(12.2)$^{⊞}$ $$\iint_{\mathcal{F}} |E_0(z;s;\chi)|^2 d\mu(z) = \frac{\mathcal{B}^{2h}}{2h} - |\varphi(s)|^2 \frac{\mathcal{B}^{-2h}}{2h} + 2\,\mathrm{Re}\left[\overline{\varphi(s)}\frac{\mathcal{B}^{2it}}{2it}\right] .$$

It follows that

$$|\varphi(s)|^2\mathcal{B}^{-2h} = \mathcal{B}^{2h} + 4h\,\mathrm{Re}\left[\overline{\varphi(s)}\frac{\mathcal{B}^{2it}}{2it}\right] - 2h\iint_{\mathcal{F}} |E_0|^2 d\mu(z)$$

$$|\varphi(s)|^2\mathcal{B}^{-2h} \leq \mathcal{B}^{2h} + 2h\left|\frac{\varphi(s)}{t}\right|$$

$$|\varphi(s)|^2 \leqq \mathcal{B}^{4\lambda} + 2\lambda \mathcal{B}^{2\lambda} \left|\frac{\varphi(s)}{t}\right| \leqq \mathcal{B}^4 + 2\mathcal{B}^2 |\varphi(s)|$$

$$2|\varphi(s)|^2 \leqq (\mathcal{B}^2 + |\varphi(s)|)^2 \quad .$$

In other words: $|\varphi(s)| \leqq (1 + \sqrt{2})B^2$. ■

Introduce $\{s_o, s_1, \ldots, s_{M_e}\}$ as in claim 9.6. The s_o entry should be omitted whenever $\zeta(\chi_o) = 0$. Define $\mathfrak{g} = \min |c|$ as in remark 8.3,⊞ and review equation (9.22).

PROPOSITION 12.5. Take $\mathrm{Re}(s) \geqq \frac{1}{2}$ and consider the analytic function

$$V(s) = \mathfrak{g}^{2s-1} \varphi(s) \prod_{k=0}^{M_e} \frac{s - s_k}{1 - s - s_k} \quad .$$

Then:

(a) $\left|V(\frac{1}{2} + iu)\right| = 1$ for $u \in \mathbb{R}$;

(b) $|V(s)| \leqq 1$;

(c) $V(\bar{s}) = \overline{V(s)}$;

(d) $\sum_{\rho} \frac{\eta}{1 + |\gamma|^2} < \infty$ where $\rho = \frac{1}{2} + \eta + i\gamma$ ranges over the zeros of $V(s)$.

It is understood here that $\eta \geqq 0$. (★)

Proof. The analyticity of $V(s)$ follows from equation (9.22) and claim 9.6. To prove assertions (a) and (c), we exploit theorem 11.8(C) and remark 8.7.

(★) The notation $\frac{1}{2} + \eta + i\gamma$ is temporary; see equation (12.6).

Proposition 8.6(d) and Stirling's formula show that V(s) is uniformly bounded over $\mathrm{Re}(s) \geqq \frac{3}{2}$. Cf. BMP[1,page 47(4)] and remark 8.3. When this fact is combined with proposition 12.4, we quickly determine that V(s) is uniformly bounded for $\mathrm{Re}(s) \geqq \frac{1}{2}$. But $|V(\frac{1}{2}+iu)| = 1$. The Phragmén-Lindelöf principle will <u>now</u> yield $|V(s)| \leqq 1$. This proves (b).

To check assertion (d), we change variables using $\xi = \frac{s-1}{s}$. Let the <u>zeros</u> of V(s) correspond to ξ_j [with the appropriate multiplicity]. Since V is uniformly bounded, the Poisson-Jensen formula shows that:

$$\sum_{\xi_j \neq 0} \ln \frac{1}{|\xi_j|} < \infty \qquad hence \qquad \sum (1-|\xi_j|^2) < \infty .$$

Cf. Ahlfors[1,page 206] and Nevanlinna[1,pages 162-165]. In other words:

$$\sum_{\rho} \frac{2\eta}{(\eta+\frac{1}{2})^2+\gamma^2} < \infty .$$

But $V(s) \neq 0$ whenever Re(s) is sufficiently large. Cf. proposition 8.6(d) and remember that $\varphi(s) \not\equiv 0$. Therefore $\sup\{\eta\} < \infty$. Assertion (d) is now obvious. ∎

The function V(s) has a meromorphic continuation to all of $\mathbb{C}$. It is apparent that $V(s)V(1-s) = 1$. Proposition 12.5 shows that the <u>poles</u> of V(s) are located at the points $1-\rho$. We also determine that $|V(s)| \geqq 1$ for $\mathrm{Re}(s) \leqq \frac{1}{2}$.

Note that $V(\frac{1}{2}) = \varphi(\frac{1}{2}) = \pm 1$. Let $\{\rho_1, \ldots, \rho_N\}$ denote those zeros of V(s) which belong to $[\frac{1}{2}, \infty)$. Recall that $\sup\{\eta\} < \infty$.

<u>PROPOSITION 12.6</u>. For $s \in \mathbb{C}$, we have:

$$V(s) = \varphi(\tfrac{1}{2})\, e^{A(s-\frac{1}{2})} \frac{(s-\rho_1)\cdots(s-\rho_N)}{(1-s-\rho_1)\cdots(1-s-\rho_N)} \prod_{\gamma>0} \frac{\left(1-\frac{s-\frac{1}{2}}{\eta+i\gamma}\right)\left(1-\frac{s-\frac{1}{2}}{\eta-i\gamma}\right)}{\left(1+\frac{s-\frac{1}{2}}{\eta+i\gamma}\right)\left(1+\frac{s-\frac{1}{2}}{\eta-i\gamma}\right)} .$$

The product is absolutely convergent whenever $s \neq \frac{1}{2} \pm \eta \pm i\gamma$. Furthermore $A \leqq 0$.

Proof. Define $s = \frac{1}{2} + \xi$ and review theorem 11.6. Apply the Hadamard factorization theorem to $F(\frac{1}{2}+\xi)$ and $G(\frac{1}{2}+\xi)$. Cf. Boas[1,pages 17 and 22]. We can certainly take $p = 4$. We immediately obtain:

$$V(s) = \exp[P(\xi)]\,\frac{(s-\rho_1)\cdots(s-\rho_N)}{(1-s-\rho_1)\cdots(1-s-\rho_N)}\;\frac{\prod\limits_{\gamma>0} E\left(\frac{\xi}{\eta+i\gamma};4\right)E\left(\frac{\xi}{\eta-i\gamma};4\right)}{\prod\limits_{\gamma>0} E\left(\frac{\xi}{-\eta+i\gamma};4\right)E\left(\frac{\xi}{-\eta-i\gamma};4\right)}$$

where $P(\xi)$ is some polynomial of degree ≤ 4.

Consider the expression

$$I_k = \left(\frac{1}{\eta+i\gamma}\right)^k + \left(\frac{1}{\eta-i\gamma}\right)^k - \left(\frac{1}{-\eta+i\gamma}\right)^k - \left(\frac{1}{-\eta-i\gamma}\right)^k$$

for $1 \leq k \leq 4$ and $\gamma \approx \infty$. Trivially $I_2 = I_4 = 0$. On the other hand:

$$I_1 = \frac{4}{R}\cos\theta \quad \text{and} \quad I_3 = \frac{4}{R^3}\cos 3\theta \qquad [\text{where } \eta+i\gamma = Re^{i\theta}]$$

Since $\gamma \approx \infty$, we see that:

$$|\cos 3\theta| \leq 6\,|\cos\theta| \quad , \qquad |I_3| \leq 6\,|I_1|$$

$$\sum_{\gamma \text{ large}} |I_1| = 4\sum_{\gamma \text{ large}} \frac{\eta}{R^2} = 4\sum_{\gamma \text{ large}} \frac{\eta}{\eta^2+\gamma^2} < \infty \qquad [\text{proposition } 12.5(c)]$$

$$\sum_{\gamma \text{ large}} |I_3| \leq 6\sum_{\gamma \text{ large}} |I_1| < \infty .$$

Therefore:

$$(*) \qquad V(s) = \exp[Q(\xi)]\cdot\frac{(s-\rho_1)\cdots(s-\rho_N)}{(1-s-\rho_1)\cdots(1-s-\rho_N)}\prod_{\gamma>0}\left[\frac{\left(1-\frac{\xi}{\eta+i\gamma}\right)\left(1-\frac{\xi}{\eta-i\gamma}\right)}{\left(1+\frac{\xi}{\eta+i\gamma}\right)\left(1+\frac{\xi}{\eta-i\gamma}\right)}\right] ,$$

where $Q(\xi)$ is a polynomial of degree ≤ 4. The product [over γ] will converge automatically whenever $s \neq \frac{1}{2} \pm \eta \pm i\gamma$. To make this fact more explicit, we take

γ very large and observe that

$$(**)\qquad \frac{\left(1-\frac{\xi}{\eta+i\gamma}\right)\left(1-\frac{\xi}{\eta-i\gamma}\right)}{\left(1+\frac{\xi}{\eta+i\gamma}\right)\left(1+\frac{\xi}{\eta-i\gamma}\right)} = 1 - \frac{4\xi\left(\frac{\eta}{\eta^2+\gamma^2}\right)}{\left(1+\frac{\xi}{\eta+i\gamma}\right)\left(1+\frac{\xi}{\eta-i\gamma}\right)} .$$

The equation $V(\frac{1}{2}+\xi)V(\frac{1}{2}-\xi) = 1$ shows that $Q(\xi) + Q(-\xi) \equiv 2n\pi i$ for some $n \in \mathbb{Z}$. Thus $Q(\xi) = n\pi i + A\xi + B\xi^3$. Using proposition 12.5(c), we establish that $A \in \mathbb{R}$ and $B \in \mathbb{R}$.

Consider the case $s \geqq 2$. Recall proposition 8.6(d) and Stirling's formula. Since $\varphi(s) \not\equiv 0$, we obtain

$$(***)\qquad |V(s)| \sim \mathcal{C}\frac{1}{\sqrt{s}}\left(\frac{\delta}{\delta_1}\right)^{2s}$$

with appropriate choices of $\mathcal{C} > 0$ and $\delta_1 \geqq \delta$. Cf. proposition 12.5.

Fix any $\delta > 0$. Note that the γ - contribution in equation (*) is sandwiched between 1 and $e^{-\delta\xi}$ for large ξ. In fact:

$$0 \leqq \frac{4\xi\left(\frac{\eta}{\eta^2+\gamma^2}\right)}{\left(1+\frac{\xi}{\eta+i\gamma}\right)\left(1+\frac{\xi}{\eta-i\gamma}\right)} = \frac{4\xi\eta}{(\eta+i\gamma+\xi)(\eta-i\gamma+\xi)} = \frac{4\xi\eta}{(\eta+\xi)^2+\gamma^2} \leqq \frac{4\eta}{\xi}$$

$$\{\text{take } \xi \geqq 4000(1+\eta)\}$$

$$-\frac{8\xi\eta}{(\eta+\xi)^2+\gamma^2} \leqq \ln\left[1 - \frac{4\xi\eta}{(\eta+\xi)^2+\gamma^2}\right] \leqq 0$$

$$-8\xi\sum_{\gamma>0}\frac{\eta}{(\eta+\xi)^2+\gamma^2} \leqq \ln\left[\prod_{\gamma>0}\right] \leqq 0 \qquad \text{cf. (*) and (**)}$$

$$-8\xi\sum_{\gamma>0}\frac{\eta}{\xi^2+\gamma^2} \leqq \ln\left[\prod_{\gamma>0}\right] \leqq 0$$

$$\left\{ \text{but} \quad \lim_{\xi \to \infty} \sum_{\gamma > 0} \frac{\eta}{\xi^2 + \gamma^2} = 0 \quad \text{via proposition 12.5(d)} \right\}$$

$$(****) \qquad e^{-\delta\xi} \leqq \prod_{\gamma > 0} \frac{\left(1 - \frac{\xi}{\eta + i\gamma}\right)\left(1 - \frac{\xi}{\eta - i\gamma}\right)}{\left(1 + \frac{\xi}{\eta + i\gamma}\right)\left(1 + \frac{\xi}{\eta - i\gamma}\right)} \leqq 1 \qquad \text{for} \qquad \xi \approx +\infty \,.$$

We combine (*) , (****) , (***) to see that $B = 0$, $A = 2\ln\left(\frac{\xi}{\xi_1}\right)$. In conclusion

$$(*****) \qquad V(s) = (-1)^n e^{A\xi} \frac{(s-\rho_1)\cdots(s-\rho_N)}{(1-s-\rho_1)\cdots(1-s-\rho_N)} \prod_{\gamma > 0} \left[\frac{\left(1 - \frac{\xi}{\eta + i\gamma}\right)\left(1 - \frac{\xi}{\eta - i\gamma}\right)}{\left(1 + \frac{\xi}{\eta + i\gamma}\right)\left(1 + \frac{\xi}{\eta - i\gamma}\right)} \right] \,.$$

Substitute $\xi = 0$ to check that $(-1)^n = V(\frac{1}{2}) = \varphi(\frac{1}{2})$. ■

NOTICE THAT:

$$(12.3) \qquad \xi_1 = \inf\left\{ x > 0 : \sum_{|c| = x} \chi_\ell(W_{00}^{-1}) \neq 0 \,,\; W_{00} \in \Gamma_\infty \backslash \Gamma / \Gamma_\infty \right\} \,.$$

Cf. proposition 8.6 and remark 8.7. One can now define

$$(12.4) \qquad V_1(s) = (\xi_1)^{2s-1} \varphi(s) \prod_{k=0}^{M_e} \frac{s - s_k}{1 - s - s_k} = \left(\frac{\xi_1}{\xi}\right)^{2s-1} V(s) \,.$$

Using (*****), we see that:

$$(12.5) \qquad V_1(s) = \varphi(\tfrac{1}{2}) \frac{(s-\rho_1)\cdots(s-\rho_N)}{(1-s-\rho_1)\cdots(1-s-\rho_N)} \prod_{\gamma > 0} \left[\frac{\left(1 - \frac{\xi}{\eta + i\gamma}\right)\left(1 - \frac{\xi}{\eta - i\gamma}\right)}{\left(1 + \frac{\xi}{\eta + i\gamma}\right)\left(1 + \frac{\xi}{\eta - i\gamma}\right)} \right] \,.$$

There is an obvious analog of proposition 12.5 for $V_1(s)$. In addition:

$$\{s_0, s_1, \ldots, s_{M_e}\} \cap \{\rho_1, \ldots, \rho_N\} = \phi \,.$$

Cf. equation (9.22).

PROPOSITION 12.7. Define:

$$\omega(r) = 1 - \frac{V_1'(\frac{1}{2}+ir)}{V_1(\frac{1}{2}+ir)} \qquad \text{for} \quad r \in \mathbb{R} .$$

Then:

(i) $\omega(r) \geqq 1$;

(ii) $\omega(r)$ is even ;

(iii) $\int_{-R}^{R} \omega(r)dr = O(R^4)$ for $R \geqq 1$; ⊞

(iv) $\omega(r) = 1 + \sum_{\rho} \frac{2\eta}{\eta^2 + (r-\gamma)^2}$ [cf. proposition 12.5(d)] .

Proof. We start with assertion (iv). First of all, by equation (12.5),

$$V_1(s) = \varphi(\tfrac{1}{2}) \frac{(s-\rho_1)\cdots(s-\rho_N)}{(1-s-\rho_1)\cdots(1-s-\rho_N)} \prod_{\gamma>0} \left[\frac{(s-\rho)(s-\bar\rho)}{(s+\rho-1)(s+\bar\rho-1)} \right] .$$

Logarithmic differentiation yields:

$$\frac{V_1'(s)}{V_1(s)} = \sum_{k=1}^{N} \left[\frac{1}{s-\rho_k} - \frac{1}{s+\rho_k-1} \right] + \sum_{\gamma>0} \left[\frac{1}{s-\rho} + \frac{1}{s-\bar\rho} - \frac{1}{s+\rho-1} - \frac{1}{s+\bar\rho-1} \right]$$

$$\frac{V_1'(\frac{1}{2}+ir)}{V_1(\frac{1}{2}+ir)} = -\sum_{k=1}^{N} \frac{2\eta_k}{\eta_k^2+r^2} - \sum_{\gamma>0} \left[\frac{2\eta}{\eta^2+(r-\gamma)^2} + \frac{2\eta}{\eta^2+(r+\gamma)^2} \right] .$$

This proves (iv), (i), (ii).

For assertion (iii), we observe that:

$$\int_{-R}^{R} \omega(r)dr = 2R + \sum_{\rho} \int_{-R}^{R} \frac{2\eta}{\eta^2+(r-\gamma)^2}\,dr \qquad [\eta>0]$$

$$\leqq 2R + \sum_{|\gamma| \leqq 2R} \int_{-\infty}^{\infty} \frac{2\eta}{\eta^2+(r-\gamma)^2}\,dr + \sum_{|\gamma|>2R} \frac{16R\eta}{|\gamma|^2}$$

$$\{ \text{exploit proposition } 12.5(d) \}$$

$$\leqq \quad 2R + 2\pi \sum_{|\gamma| \leqq 2R} 1 + o(R) \quad .$$

The zeros of $V(s)$ correspond to the zeros of $\varphi(s)$ whenever $\mathrm{Re}(s) \geqq \frac{1}{2}$. Apply theorem 11.6 and Boas[1,page 16(2.5.13)]. Study the zeros of $F(s)$. We immediately obtain:

$$\sum_{|\gamma| \leqq 2R} 1 = O(R^4) .$$

Assertion (iii) is now obvious. ■

PROPOSITION 12.8. Let $s = \sigma + it$. For $\frac{1}{2} \leqq \sigma \leqq \frac{3}{2}$ and $|t| \geqq 1$, we have:

$$1 - |\varphi(s)|^2 \leqq O[(\sigma - \tfrac{1}{2})\,\omega(t)] .$$

The implied constant will depend solely on Γ, χ.

Proof. Recall equation (12.4). Hence:

$$
\begin{aligned}
1 - |\varphi(s)|^2 &= 1 - |V_1(s)|^2 + |V_1(s)|^2 \left[1 - \left| \frac{\varphi(s)}{V_1(s)} \right|^2 \right] \\
&= 1 - |V_1(s)|^2 + |V_1(s)|^2 \left[1 - (g_1)^{2-4\sigma} \prod_{k=0}^{M} \left| \frac{s + s_k - 1}{s - s_k} \right|^2 \right] \\
&\leqq 1 - |V_1(s)|^2 + |V_1(s)|^2 \left[1 - (g_1)^{2-4\sigma} \right] \\
&\leqq 1 - |V_1(s)|^2 + O(\sigma - \tfrac{1}{2}) \qquad [\text{since } |V_1(s)| \leqq 1] .
\end{aligned}
$$

But,

$$1-|V_1(s)|^2 = 1 - \left|\frac{(s-\rho_1)\cdots(s-\rho_N)}{(s+\rho_1-1)\cdots(s+\rho_N-1)}\right|^2 \cdot \prod_{\gamma>0}\left|\frac{(s-\rho)(s-\bar\rho)}{(s+\bar\rho-1)(s+\rho-1)}\right|^2$$

$$= 1 - \prod_{j=1}^{N}\left[1-\frac{4\eta_j(\sigma-\frac{1}{2})}{(\sigma-\frac{1}{2}+\eta_j)^2+t^2}\right]\cdot\prod_{\gamma\neq 0}\left[1-\frac{4\eta(\sigma-\frac{1}{2})}{(\sigma-\frac{1}{2}+\eta)^2+(t-\gamma)^2}\right]$$

$$= 1 - \prod_{\rho}\left[1-\frac{4\eta(\sigma-\frac{1}{2})}{(\sigma-\frac{1}{2}+\eta)^2+(t-\gamma)^2}\right] \qquad \{\eta>0\}$$

$$\leqq \sum_{\rho}\frac{4\eta(\sigma-\frac{1}{2})}{(\sigma-\frac{1}{2}+\eta)^2+(t-\gamma)^2} \qquad \{\text{easily enough}\}$$

$$\leqq 2(\sigma-\tfrac{1}{2})\sum_{\rho}\frac{2\eta}{\eta^2+(t-\gamma)^2}$$

$$\leqq 2(\sigma-\tfrac{1}{2})\,\omega(t) \qquad \{\text{cf. proposition } 12.7(\text{iv})\} .$$

Therefore $1-|\varphi(s)|^2 \leqq O[(\sigma-\frac{1}{2})\,\omega(t)]$. ■

DEFINE:

(12.6) $$\eta = \frac{1}{20}\inf\{\mathrm{Im}(z) : z \in \mathfrak{F}\} .$$

Recall (ix) in section 3. By increasing the value of y_0 , we can ensure that:

(12.7) $$\mathrm{Im}\; T(z) \leqq \frac{\eta}{2} \quad \text{for } T \in \Gamma - [S] \text{ and } y \geqq y_0 .$$ ⊞

Cf. the proof of proposition 8.1. This modification involves no loss of generality.

Let δ be a <u>tiny</u> number so that the disks $|s-a| \leqq \delta$ are disjoint for $a \in \{\frac{1}{2}, 1, s_1, \ldots, s_{M_e}\}$. Write $s = \sigma + it$.

THEOREM 12.9. Take $\frac{1}{2} \leqq \sigma \leqq \frac{3}{2}$ but keep $|s-s_k| \geqq \delta$ for $0 \leqq k \leqq M_e$. Recall the convention about s_o [page 156]. Then:

(a) $\varphi(s)$ is uniformly bounded;

(b) $1-|\varphi(s)|^2 \leqq O[(\sigma-\frac{1}{2})\omega(t)]$;

(c) $\varphi_m(s) = O[\sqrt{\omega(t)}\, e^{3|t|+5\pi|m|\eta}]$ for $m \neq 0$;

(d) $E(z;s;\mathcal{X}) = y^s + \varphi(s)y^{1-s} + O[\sqrt{\omega(t)}\, e^{3|t|-2\pi y}]$ for $y \geqq 10\eta$.

The implied constants depend solely on Γ, $\mathcal{X}$, $\mathfrak{F}$, δ. They are independent of m, σ, t. ⊞

Proof. It is easily seen that assertions (a)-(d) are valid when $|t| \leqq 1$. For (c) and (d), one uses the analog of (9.19). See also proposition 12.7(i).

We may now assume WLOG that $|t| \geqq 1$. Assertions (a) and (b) correspond to propositions 12.4 and 12.8.

To prove (c), we exploit equation (12.2). For $0 \leqq x \leqq 1$, we define:

$$E_{00}(z;s;\mathcal{X}) = \begin{cases} E(z;s;\mathcal{X}) & \text{when } 0<y<B \\ E(z;s;\mathcal{X}) - y^s - \varphi(s)y^{1-s} & \text{when } B<y<\infty \end{cases} .$$

Cf. page 154 top. Review equations (12.1) and (12.7). Therefore

$$[0,1]\times[\eta,y_0] \subseteqq T_1(\mathfrak{F}_B) \cup \cdots \cup T_K(\mathfrak{F}_B)$$

for <u>some</u> choice of $\{T_1,\ldots,T_k\} \subseteqq \Gamma$. Furthermore [by (12.2)]:

$$\int_\eta^\infty \int_0^1 |E_{00}(z;s;\chi)|^2 \frac{dx\,dy}{y^2} = \int_\eta^{y_0}\int_0^1 + \int_{y_0}^\infty \int_0$$

$$\leqq k \int_{\mathcal{F}_B} |E(z;s;\chi)|^2 d\mu(z) + \int_{C_{y_0}} |E_0(z;s;\chi)|^2 d\mu(z)$$

$$\leqq (k+1) \int_{\mathcal{F}} |E_0(z;s;\chi)|^2 d\mu(z)$$

$$= (k+1)\left[\frac{B^{2\sigma-1} - B^{1-2\sigma}}{2\sigma-1} + (1-|\varphi(s)|^2)\frac{B^{1-2\sigma}}{2\sigma-1} \right]$$

$$+ (2k+2)\,\mathrm{Re}\left[\overline{\varphi(s)}\, \frac{B^{2it}}{2it} \right]$$

$$\left\{ \frac{\text{recall assertions (a) and (b)}}{\text{WLOG } \tfrac{1}{2} < \sigma \leqq \tfrac{3}{2}} \right\}$$

$$\leqq O(1) + O[\omega(t)]$$

$$= O[\omega(t)] \quad .$$

We can now exploit proposition 8.6 and Parseval's formula. Thus:

$$\sum_{m\neq 0} \int_\eta^\infty |\varphi_m(s) y^{\frac{1}{2}} K_{s-\frac{1}{2}}(2\pi|m|y)|^2 \frac{dy}{y^2} = O[\omega(t)]$$

$$\sum_{m\neq 0} |\varphi_m(s)|^2 \int_\eta^\infty |K_{s-\frac{1}{2}}(2\pi|m|y)|^2 \frac{dy}{y} = O[\omega(t)]$$

$$\sum_{m \neq 0} |\varphi_m(s)|^2 \int_{2\pi|m|/\eta}^{\infty} |K_{s-\frac{1}{2}}(y)|^2 \frac{dy}{y} \quad = \quad O[w(t)]$$

{ consult proposition 12.2 }

$$A \sum_{m \neq 0} |\varphi_m(s)|^2 e^{-5|t| - 10\pi|m|/\eta} \quad = \quad O[w(t)] \quad .$$

Assertion (c) follows immediately. Assertion (d) is a trivial consequence of (c) and propositions 8.6, 12.1. ■

Theorem 12.9 should be compared with page 76 line 10. Our estimate for $\varphi_m(s)$ is not that bad. See Titchmarsh[4, pages 44(line 11) + 137(line 19) + 165(top)] and 227 note 131.

For later reference, we note that:

(12.8) $$\varphi(s) = (\zeta_1)^{1-2s} \varphi(\tfrac{1}{2}) \prod_{k=0}^{M_e} \frac{1-s-s_k}{s-s_k} \cdot \frac{(s-\rho_1)\cdots(s-\rho_N)}{(1-s-\rho_1)\cdots(1-s-\rho_N)} \prod_{\gamma>0} \frac{(s-\rho)(s-\bar{\rho})}{(s+\rho-1)(s+\bar{\rho}-1)} \quad .$$

This identity follows from equation (12.5). Do not confuse (12.6) and the [earlier] notation $\rho = \frac{1}{2} + \eta + i\gamma$. As usual: $k = 0$ is omitted whenever $\delta(\chi) = 0$.

13. Development of the trace formula when $\chi(S) = 1$.

We begin with some informal remarks. Suppose that $k(z,w)$ is a reasonable point-pair invariant. Form

(13.1) $$K(z,w,\chi) = \sum_{T \in \Gamma} \chi(T)\, k(z, Tw)$$

as in section 9. See page 67. Hold w fixed so that $K(z,w,\chi) \in L_2(\Gamma \backslash H, \chi)$.